INDIAN SPACE OLYMPIAD

2025 MOCK QUESTIONS

GRADE 4

FOUNDATION LEVEL

5 MOCK PAPERS | 50 QUESTIONS EACH

Based on the ISO Foundation Level syllabus

ISO 2025 Mock Questions - Grade 4 - Foundation Level

About These Grade 4 Mock Papers

Purpose. These five papers are written only for Grade 4 learners. They use the Indian Space Olympiad Foundation Level syllabus, but the wording, calculations and reasoning have been kept suitable for children of this grade.

Difficulty. Questions 1-40 in every paper are mostly easy, with a few moderate questions. Questions 41-50 form the Olympiad Challenge. These are harder, but they remain within the Foundation Level syllabus and the expected ability of the grade.

Suggested use. A parent or teacher may read a question aloud when needed. Encourage the child to explain why an option is correct instead of guessing quickly.

Paper	Questions	Suggested Time	Marks
1	50	75 minutes	50
2	50	75 minutes	50
3	50	75 minutes	50
4	50	75 minutes	50
5	50	75 minutes	50

Syllabus Areas Used

Fundamentals of Astronomy

The Solar System

Stars, Galaxies and the Universe

Earth and Moon

Satellites and Space Technology

Rockets and Launch Vehicles

ISRO and India's Space Missions

Space Exploration

Basic Physics for Space Science

Scientific Reasoning and Logical Aptitude

General Space Awareness

Mock Paper 1

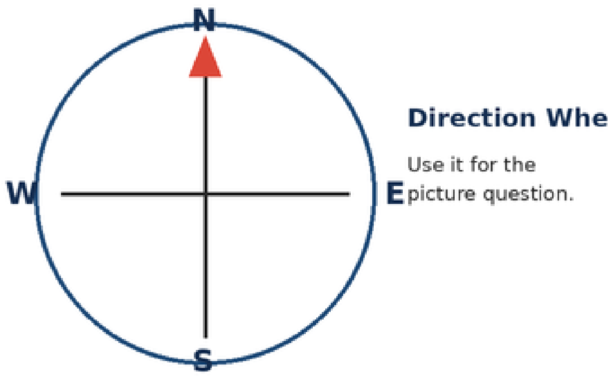
2025 Mock Questions - Grade 4 - Foundation Level

Questions	Marks	Time	Negative marking
50	50	75 minutes	None

Instructions: Choose one correct option for each question. Questions 41-50 are the Olympiad Challenge. Work carefully and use the pictures where shown.

Section A - Core Questions

1. On the direction wheel, which direction lies opposite East?



A. North	B. South
C. Up	D. West

2. A good astronomical observation record should include:

A. date, time and what was seen	B. only the observer's favourite colour
C. only the name of a planet	D. a guess with no observation

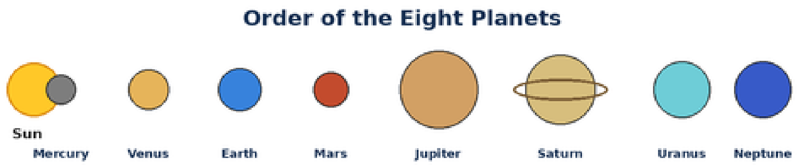
3. Earth has seasons mainly because it:

A. moves closer to the Moon every winter	B. revolves around the Sun with a tilted axis
C. stops rotating in summer	D. changes its size each month

4. Constellations help sky-watchers to:

A. make stars move into new places	B. turn planets into stars
C. recognise patterns and regions in the night sky	D. measure ocean depth directly

5. Which list gives the first four planets in the correct order from the Sun?



A. Venus, Mercury, Mars, Earth	B. Earth, Venus, Mercury, Mars
C. Mars, Earth, Venus, Mercury	D. Mercury, Venus, Earth, Mars

6. Which pair contains two rocky inner planets?

A. Earth and Mars	B. Jupiter and Saturn
C. Saturn and Neptune	D. Jupiter and Uranus

7. Pluto is classified as a:

A. star	B. dwarf planet
C. natural satellite of Earth	D. comet tail

8. Which statement best separates a comet from an asteroid?

A. A comet is always a planet	B. An asteroid is made only of water
C. A comet contains much ice and can form a tail near the Sun	D. An asteroid shines by making its own light

9. Which planet is nearest to the Sun?

A. Venus	B. Earth
C. Mars	D. Mercury

10. The Sun is important to Earth because it provides:

A. light and heat	B. rings and craters
C. ocean salt only	D. all of Earth's rocks

11. Which statement is correct?

A. The Milky Way contains the whole Universe	B. The Universe contains many galaxies
C. Every star is a galaxy	D. Earth is larger than every galaxy

12. Two stars may look different in brightness because they can differ in:

A. number of oceans	B. amount of soil
C. distance and actual brightness	D. type of rocket stage

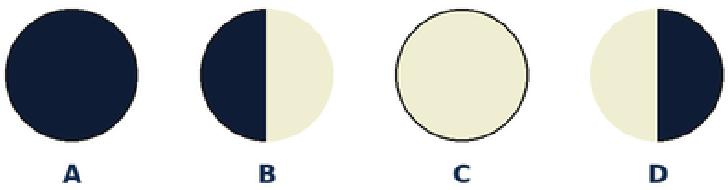
13. About how long does Earth take to rotate once?

A. one week	B. one month
C. 365 days	D. 24 hours

14. About how long does Earth take to revolve once around the Sun?

A. 365 days	B. 24 hours
C. seven days	D. one lunar night

15. In the Moon drawings, which phase normally comes after B as the lit part grows?



Four simple Moon-phase drawings

A. A	B. C
C. B again	D. D

16. During a lunar eclipse, which object is between the Sun and Moon?

A. Mars	B. Venus
C. Earth	D. a comet

17. Ocean tides are mainly linked with the pull of the:

A. North Star	B. Milky Way
C. weather satellite	D. Moon

18. A satellite that carries television and phone signals is a:

A. communication satellite	B. weather balloon only
C. natural satellite	D. dwarf planet

19. A weather satellite is most useful for observing:

A. the inside of a school bag	B. clouds, storms and large weather systems
C. underground train seats	D. the colour of one leaf indoors

20. GPS satellites help a navigation app find:

A. the age of the Sun	B. the taste of water
C. location and route	D. the number of stars in the Universe

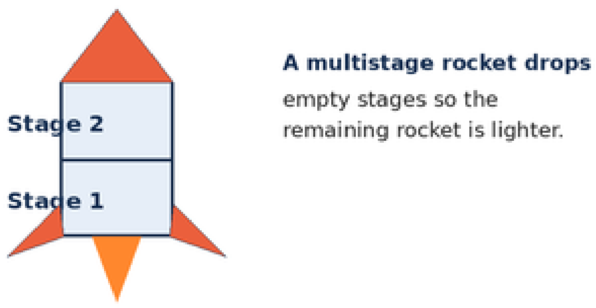
21. Which pair contains one natural and one artificial satellite?

A. Earth and Mars	B. Sun and Jupiter
C. comet and asteroid	D. Moon and weather satellite

22. A rocket rises because it pushes hot gases downward and the gases push the rocket:

A. upward	B. into the launch pad
C. towards Earth's centre only	D. without any motion

23. Why do many launch vehicles use stages?



A. Stages make gravity disappear	B. Empty parts can be dropped to reduce mass
C. Each stage becomes a planet	D. Stages stop the engine from working

24. A reusable launch vehicle is designed to:

A. remain on the Moon forever	B. turn into a constellation
C. use some major parts again	D. stop after one second

25. The full form of ISRO is:

A. International Star Reading Office	B. Indian Sky Rocket Observatory
C. Institute of Solar Ring Operations	D. Indian Space Research Organisation

26. The Vikram lander and Pragyan rover were parts of:

A. Chandrayaan-3	B. Mangalyaan
C. Aditya-L1	D. Gaganyaan

27. India's Mars Orbiter Mission is popularly called:

A. Chandrayaan	B. Mangalyaan
C. Gaganyaan	D. Aryabhata only

28. Aditya-L1 is an Indian mission designed to observe the:

A. deep ocean	B. rings of Saturn only
C. Sun	D. surface of Mars

29. Gaganyaan aims to develop India's ability for:

A. moving the Moon	B. building a city on the Sun
C. stopping Earth's revolution	D. human spaceflight

30. Astronauts inside an orbiting spacecraft appear to float because they and the spacecraft are:

A. falling around Earth together	B. outside all gravity
C. being pushed by sunlight only	D. standing on a hidden floor

31. The International Space Station travels mainly in orbit around:

A. Mars	B. Earth
C. the Sun at L1	D. Jupiter

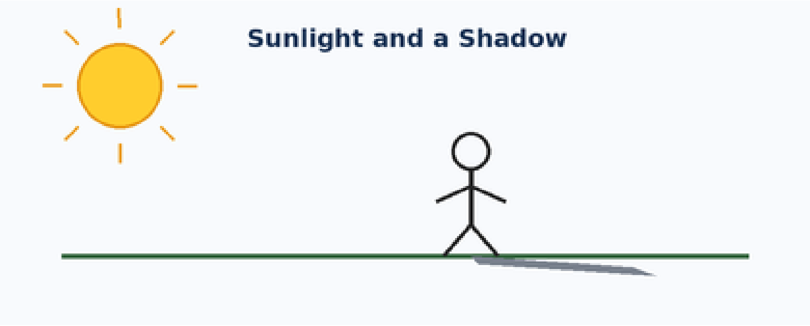
32. A space telescope can observe from above much of Earth's:

A. core	B. ocean floor
C. atmosphere	D. soil

33. If a moving toy car is pushed harder in the same direction, it will usually:

A. turn into a magnet	B. lose all mass
C. make darkness	D. speed up

34. A clear shadow forms when:



A. an opaque object blocks light	B. sound passes through air
C. heat reaches water	D. gravity pulls a stone

35. Which object changes electrical energy mainly into light?

A. a wooden ruler	B. an electric bulb
C. a stone	D. a paper clip alone

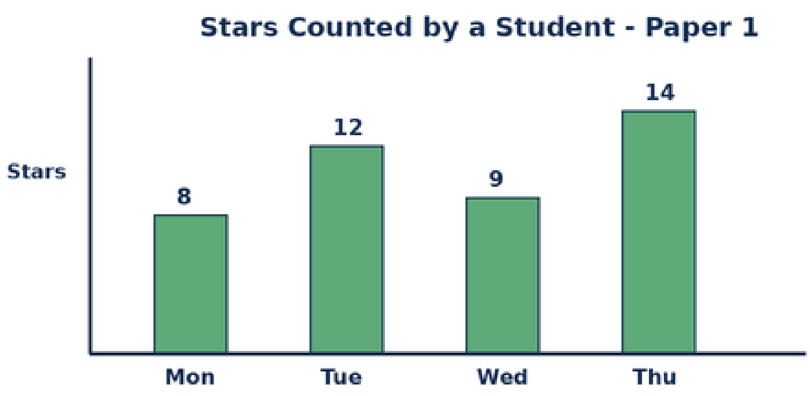
36. A ramp is an example of which simple machine?

A. pulley	B. wheel only
C. inclined plane	D. thermometer

37. Earth's atmosphere is important because it provides air and also helps:

A. stop all sunlight	B. remove gravity
C. make Earth square	D. protect life and control temperature

38. Use the bar chart. What is the difference between the highest and lowest star counts?



A. 6	B. 7
C. 14	D. 43

39. Complete the number pattern: 3, 5, 7, ___.

A. 8	B. 9
C. 11	D. 7

40. Which mission-purpose match is correct?

A. Mangalyaan - study of the Sun	B. Aditya-L1 - human flight around Earth
C. Chandrayaan - Moon exploration	D. Gaganyaan - study of Saturn's rings

Section B - Olympiad Challenge

These ten questions are harder, but every question remains within the Grade-level Foundation syllabus.

41. Using the planet order, how many planets lie between Venus and Jupiter?

A. 1	B. 3
C. 4	D. 2

42. Which alignment can produce a solar eclipse?

A. Sun - Moon - Earth	B. Sun - Earth - Moon
C. Moon - Sun - Earth	D. Earth - Sun - Moon

43. Which pair correctly matches Earth's motions and times?

A. rotation - 365 days; revolution - 24 hours	B. rotation - about 24 hours; revolution - about 365 days
C. rotation - one month; revolution - one week	D. rotation - one year; revolution - one day

44. Arjun checks a cyclone map and then follows a route to a shelter. Which two satellite uses are involved?

A. communication and Moon exploration	B. tides and rocket staging
C. weather observation and navigation	D. constellations and simple machines

45. A rocket has used all the fuel in its first stage. What is the best next action in a multistage design?

A. Carry the empty stage forever	B. Turn off every remaining engine
C. Open the astronaut's helmet	D. Separate the empty stage

46. The four daily star counts are 8, 12, 9 and 14. What is their total?

A. 43	B. 41
C. 45	D. 14

47. Which statement correctly compares a star, a planet and a moon?

A. All three make their own light	B. A star makes light; a planet orbits a star; a moon orbits a planet
C. A planet orbits a moon	D. A moon is always larger than a star

48. The sequence new Moon -> first quarter -> full Moon takes about:

A. one hour	B. one day
C. two weeks	D. one year

49. Two children push a box with equal force in opposite directions. What is most likely if the box was at rest?

A. It moves quickly in both directions	B. It flies upward
C. It loses gravity	D. It remains at rest

50. All planets in our Solar System orbit the Sun. Earth is a planet. Which conclusion follows?

A. Earth orbits the Sun	B. Earth is the Sun
C. The Sun orbits Earth every day	D. Earth is a natural satellite

Mock Paper 2

2025 Mock Questions - Grade 4 - Foundation Level

Questions	Marks	Time	Negative marking
50	50	75 minutes	None

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Section A - Core Questions

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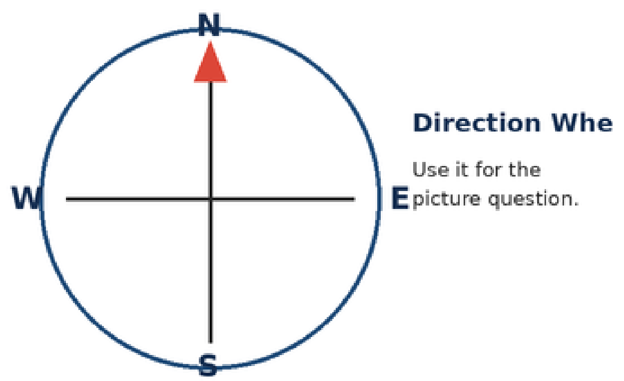

A. North	B. West
C. South	D. Up

2. A good astronomical observation record should include:

A. only the observer's favourite colour	B. only the name of a planet
C. date, time and what was seen	D. a guess with no observation

3. Earth has seasons mainly because it:

A. moves closer to the Moon every winter	B. stops rotating in summer
C. changes its size each month	D. revolves around the Sun with a tilted axis

4. Constellations help sky-watchers to:

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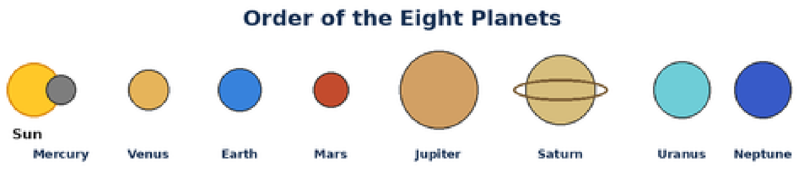

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C. one month	D. 365 days

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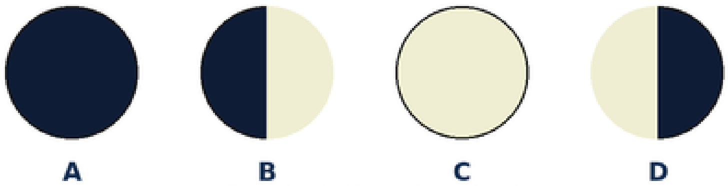

Four simple Moon-phase drawings

A. A	B. B again
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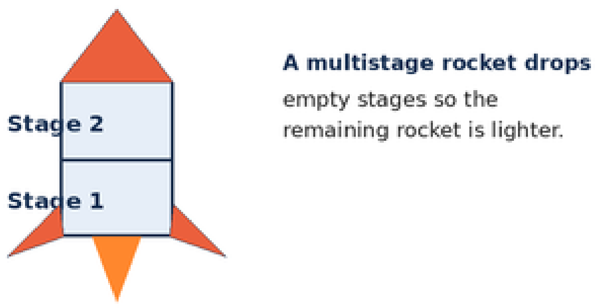

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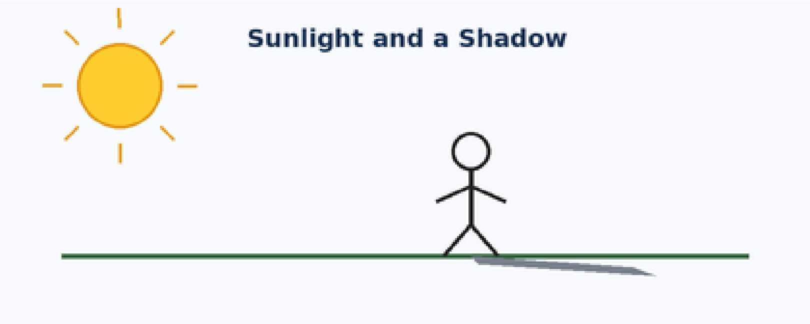

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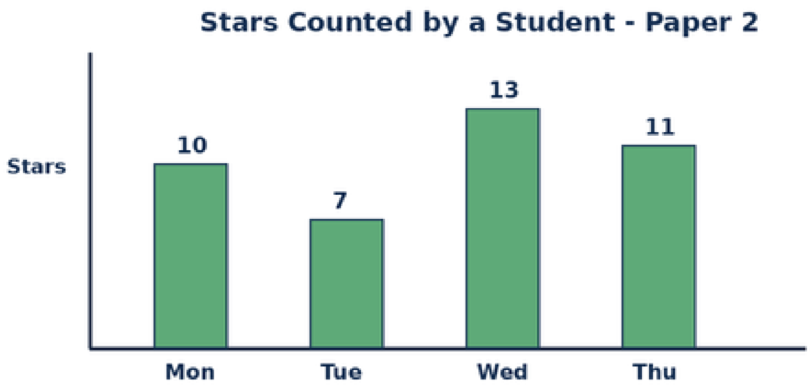

A. 7	B. 13
C. 6	D. 41

39. Complete the number pattern: 4, 6, 8, ___.

A. 9	B. 12
C. 8	D. 10

40. Which mission-purpose match is correct?

A. Chandrayaan - Moon exploration	B. Mangalyaan - study of the Sun
C. Aditya-L1 - human flight around Earth	D. Gaganyaan - study of Saturn's rings

Section B - Olympiad Challenge

These ten questions are harder, but every question remains within the Grade-level Foundation syllabus.

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A. 1	B. 2
C. 3	D. 4

42. Which alignment can produce a solar eclipse?

A. Sun - Earth - Moon	B. Moon - Sun - Earth
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43. Which pair correctly matches Earth's motions and times?

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C. rotation - one year; revolution - one day	D. rotation - about 24 hours; revolution - about 365 days

44. Rohan checks a cyclone map and then follows a route to a shelter. Which two satellite uses are involved?

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45. A rocket has used all the fuel in its first stage. What is the best next action in a multistage design?

A. Carry the empty stage forever	B. Separate the empty stage
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46. The four daily star counts are 10, 7, 13 and 11. What is their total?

A. 39	B. 43
C. 41	D. 13

47. Which statement correctly compares a star, a planet and a moon?

A. All three make their own light	B. A planet orbits a moon
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48. The sequence new Moon -> first quarter -> full Moon takes about:

A. two weeks	B. one hour
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49. Two children push a box with equal force in opposite directions. What is most likely if the box was at rest?

A. It moves quickly in both directions	B. It remains at rest
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A. Earth is the Sun	B. The Sun orbits Earth every day
C. Earth orbits the Sun	D. Earth is a natural satellite

Mock Paper 3

2025 Mock Questions - Grade 4 - Foundation Level

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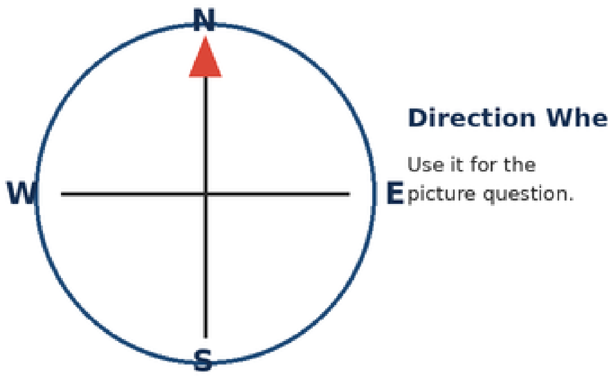

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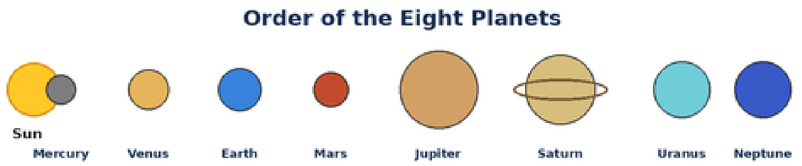

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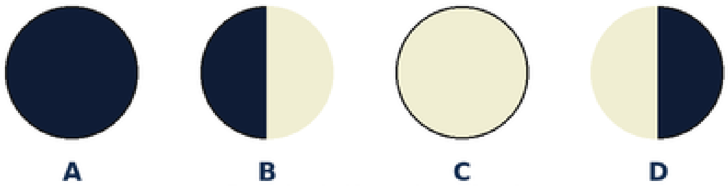

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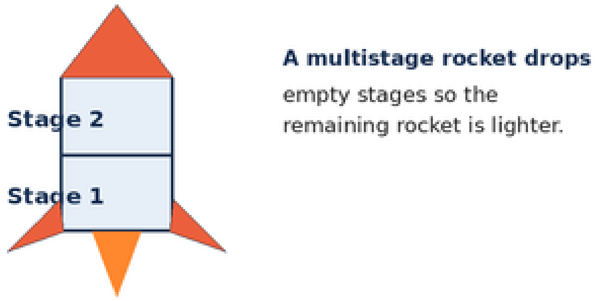

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A. deep ocean	B. rings of Saturn only
C. Sun	D. surface of Mars

29. Gaganyaan aims to develop India's ability for:

A. moving the Moon	B. building a city on the Sun
C. stopping Earth's revolution	D. human spaceflight

30. Astronauts inside an orbiting spacecraft appear to float because they and the spacecraft are:

A. falling around Earth together	B. outside all gravity
C. being pushed by sunlight only	D. standing on a hidden floor

31. The International Space Station travels mainly in orbit around:

A. Mars	B. Earth
C. the Sun at L1	D. Jupiter

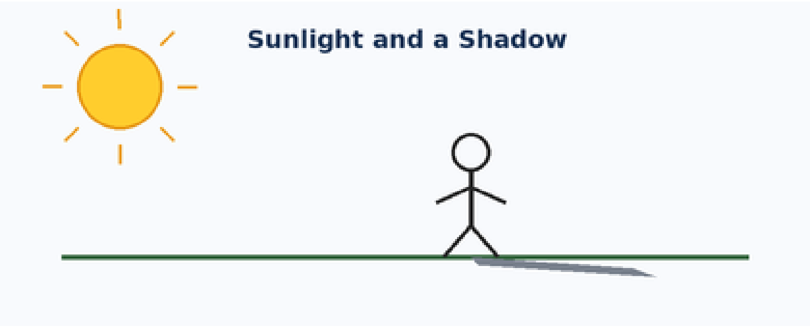
32. A space telescope can observe from above much of Earth's:

A. core	B. ocean floor
C. atmosphere	D. soil

33. If a moving toy car is pushed harder in the same direction, it will usually:

A. turn into a magnet	B. lose all mass
C. make darkness	D. speed up

34. A clear shadow forms when:



A. an opaque object blocks light	B. sound passes through air
C. heat reaches water	D. gravity pulls a stone

35. Which object changes electrical energy mainly into light?

A. a wooden ruler	B. an electric bulb
C. a stone	D. a paper clip alone

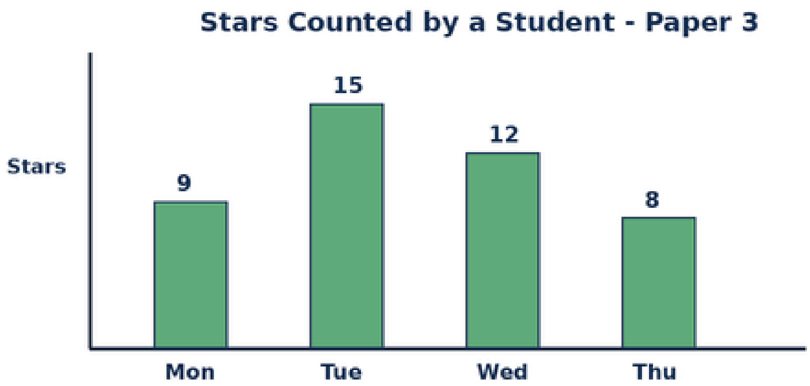
36. A ramp is an example of which simple machine?

A. pulley	B. wheel only
C. inclined plane	D. thermometer

37. Earth's atmosphere is important because it provides air and also helps:

A. stop all sunlight	B. remove gravity
C. make Earth square	D. protect life and control temperature

38. Use the bar chart. What is the difference between the highest and lowest star counts?



A. 7	B. 8
C. 15	D. 44

39. Complete the number pattern: 5, 7, 9, ___.

A. 10	B. 11
C. 13	D. 9

40. Which mission-purpose match is correct?

A. Mangalyaan - study of the Sun	B. Aditya-L1 - human flight around Earth
C. Chandrayaan - Moon exploration	D. Gaganyaan - study of Saturn's rings

Section B - Olympiad Challenge

These ten questions are harder, but every question remains within the Grade-level Foundation syllabus.

41. Using the planet order, how many planets lie between Earth and Saturn?

A. 1	B. 3
C. 4	D. 2

42. Which alignment can produce a solar eclipse?

A. Sun - Moon - Earth	B. Sun - Earth - Moon
C. Moon - Sun - Earth	D. Earth - Sun - Moon

43. Which pair correctly matches Earth's motions and times?

A. rotation - 365 days; revolution - 24 hours	B. rotation - about 24 hours; revolution - about 365 days
C. rotation - one month; revolution - one week	D. rotation - one year; revolution - one day

44. Aarav checks a cyclone map and then follows a route to a shelter. Which two satellite uses are involved?

A. communication and Moon exploration	B. tides and rocket staging
C. weather observation and navigation	D. constellations and simple machines

45. A rocket has used all the fuel in its first stage. What is the best next action in a multistage design?

A. Carry the empty stage forever	B. Turn off every remaining engine
C. Open the astronaut's helmet	D. Separate the empty stage

46. The four daily star counts are 9, 15, 12 and 8. What is their total?

A. 44	B. 42
C. 46	D. 15

47. Which statement correctly compares a star, a planet and a moon?

A. All three make their own light	B. A star makes light; a planet orbits a star; a moon orbits a planet
C. A planet orbits a moon	D. A moon is always larger than a star

48. The sequence new Moon -> first quarter -> full Moon takes about:

A. one hour	B. one day
C. two weeks	D. one year

49. Two children push a box with equal force in opposite directions. What is most likely if the box was at rest?

A. It moves quickly in both directions	B. It flies upward
C. It loses gravity	D. It remains at rest

50. All planets in our Solar System orbit the Sun. Earth is a planet. Which conclusion follows?

A. Earth orbits the Sun	B. Earth is the Sun
C. The Sun orbits Earth every day	D. Earth is a natural satellite

Mock Paper 4

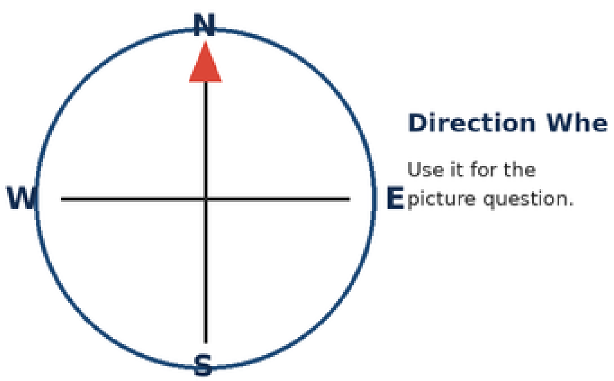
2025 Mock Questions - Grade 4 - Foundation Level

Questions	Marks	Time	Negative marking
50	50	75 minutes	None

Instructions: Choose one correct option for each question. Questions 41-50 are the Olympiad Challenge. Work carefully and use the pictures where shown.

Section A - Core Questions

1. On the direction wheel, which direction lies opposite East?



A. North	B. West
C. South	D. Up

2. A good astronomical observation record should include:

A. only the observer's favourite colour	B. only the name of a planet
C. date, time and what was seen	D. a guess with no observation

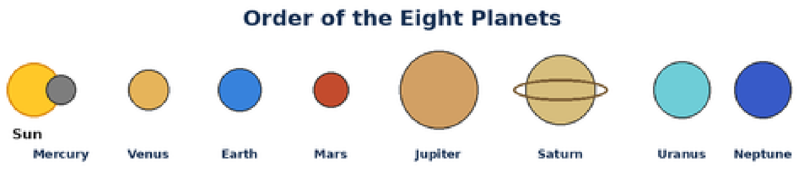
3. Earth has seasons mainly because it:

A. moves closer to the Moon every winter	B. stops rotating in summer
C. changes its size each month	D. revolves around the Sun with a tilted axis

4. Constellations help sky-watchers to:

A. recognise patterns and regions in the night sky	B. make stars move into new places
C. turn planets into stars	D. measure ocean depth directly

5. Which list gives the first four planets in the correct order from the Sun?



A. Venus, Mercury, Mars, Earth	B. Mercury, Venus, Earth, Mars
C. Earth, Venus, Mercury, Mars	D. Mars, Earth, Venus, Mercury

6. Which pair contains two rocky inner planets?

A. Jupiter and Saturn	B. Saturn and Neptune
C. Earth and Mars	D. Jupiter and Uranus

7. Pluto is classified as a:

A. star	B. natural satellite of Earth
C. comet tail	D. dwarf planet

8. Which statement best separates a comet from an asteroid?

A. A comet contains much ice and can form a tail near the Sun	B. A comet is always a planet
C. An asteroid is made only of water	D. An asteroid shines by making its own light

9. Which planet is nearest to the Sun?

A. Venus	B. Mercury
C. Earth	D. Mars

10. The Sun is important to Earth because it provides:

A. rings and craters	B. ocean salt only
C. light and heat	D. all of Earth's rocks

11. Which statement is correct?

A. The Milky Way contains the whole Universe	B. Every star is a galaxy
C. Earth is larger than every galaxy	D. The Universe contains many galaxies

12. Two stars may look different in brightness because they can differ in:

A. distance and actual brightness	B. number of oceans
C. amount of soil	D. type of rocket stage

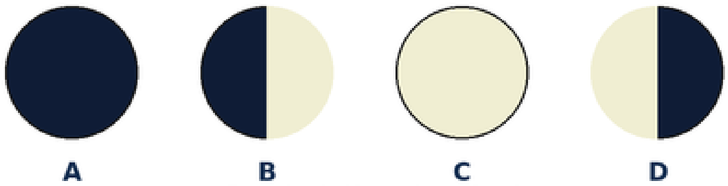
13. About how long does Earth take to rotate once?

A. one week	B. 24 hours
C. one month	D. 365 days

14. About how long does Earth take to revolve once around the Sun?

A. 24 hours	B. seven days
C. 365 days	D. one lunar night

15. In the Moon drawings, which phase normally comes after B as the lit part grows?



Four simple Moon-phase drawings

A. A	B. B again
C. D	D. C

16. During a lunar eclipse, which object is between the Sun and Moon?

A. Earth	B. Mars
C. Venus	D. a comet

17. Ocean tides are mainly linked with the pull of the:

A. North Star	B. Moon
C. Milky Way	D. weather satellite

18. A satellite that carries television and phone signals is a:

A. weather balloon only	B. natural satellite
C. communication satellite	D. dwarf planet

19. A weather satellite is most useful for observing:

A. the inside of a school bag	B. underground train seats
C. the colour of one leaf indoors	D. clouds, storms and large weather systems

20. GPS satellites help a navigation app find:

A. location and route	B. the age of the Sun
C. the taste of water	D. the number of stars in the Universe

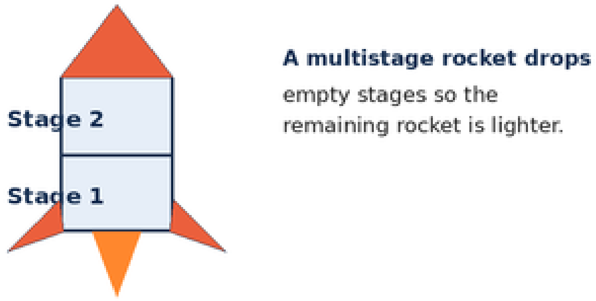
21. Which pair contains one natural and one artificial satellite?

A. Earth and Mars	B. Moon and weather satellite
C. Sun and Jupiter	D. comet and asteroid

22. A rocket rises because it pushes hot gases downward and the gases push the rocket:

A. into the launch pad	B. towards Earth's centre only
C. upward	D. without any motion

23. Why do many launch vehicles use stages?



A. Stages make gravity disappear	B. Each stage becomes a planet
C. Stages stop the engine from working	D. Empty parts can be dropped to reduce mass

24. A reusable launch vehicle is designed to:

A. use some major parts again	B. remain on the Moon forever
C. turn into a constellation	D. stop after one second

25. The full form of ISRO is:

A. International Star Reading Office	B. Indian Space Research Organisation
C. Indian Sky Rocket Observatory	D. Institute of Solar Ring Operations

26. The Vikram lander and Pragyan rover were parts of:

A. Mangalyaan	B. Aditya-L1
C. Chandrayaan-3	D. Gaganyaan

27. India's Mars Orbiter Mission is popularly called:

A. Chandrayaan	B. Gaganyaan
C. Aryabhata only	D. Mangalyaan

28. Aditya-L1 is an Indian mission designed to observe the:

A. Sun	B. deep ocean
C. rings of Saturn only	D. surface of Mars

29. Gaganyaan aims to develop India's ability for:

A. moving the Moon	B. human spaceflight
C. building a city on the Sun	D. stopping Earth's revolution

30. Astronauts inside an orbiting spacecraft appear to float because they and the spacecraft are:

A. outside all gravity	B. being pushed by sunlight only
C. falling around Earth together	D. standing on a hidden floor

31. The International Space Station travels mainly in orbit around:

A. Mars	B. the Sun at L1
C. Jupiter	D. Earth

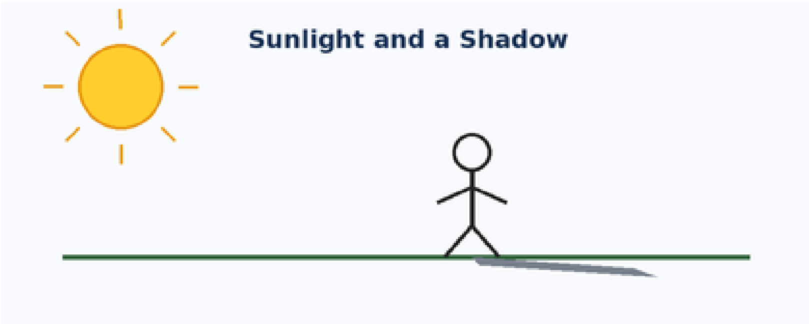
32. A space telescope can observe from above much of Earth's:

A. atmosphere	B. core
C. ocean floor	D. soil

33. If a moving toy car is pushed harder in the same direction, it will usually:

A. turn into a magnet	B. speed up
C. lose all mass	D. make darkness

34. A clear shadow forms when:



A. sound passes through air	B. heat reaches water
C. an opaque object blocks light	D. gravity pulls a stone

35. Which object changes electrical energy mainly into light?

A. a wooden ruler	B. a stone
C. a paper clip alone	D. an electric bulb

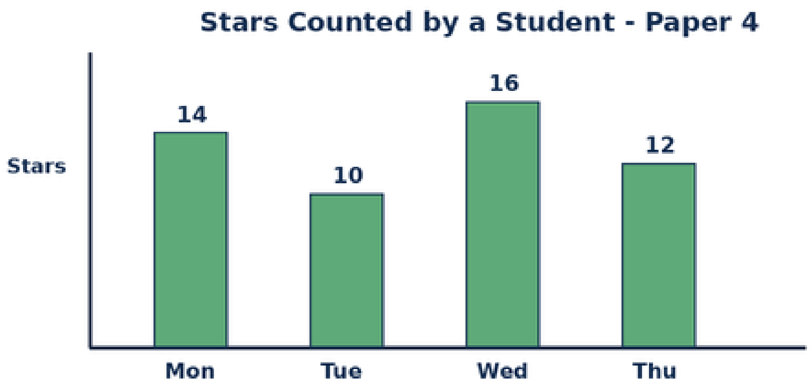
36. A ramp is an example of which simple machine?

A. inclined plane	B. pulley
C. wheel only	D. thermometer

37. Earth's atmosphere is important because it provides air and also helps:

A. stop all sunlight	B. protect life and control temperature
C. remove gravity	D. make Earth square

38. Use the bar chart. What is the difference between the highest and lowest star counts?



A. 7	B. 16
C. 6	D. 52

39. Complete the number pattern: 6, 8, 10, ___.

A. 11	B. 14
C. 10	D. 12

40. Which mission-purpose match is correct?

A. Chandrayaan - Moon exploration	B. Mangalyaan - study of the Sun
C. Aditya-L1 - human flight around Earth	D. Gaganyaan - study of Saturn's rings

Section B - Olympiad Challenge

These ten questions are harder, but every question remains within the Grade-level Foundation syllabus.

41. Using the planet order, how many planets lie between Mars and Uranus?

A. 2	B. 3
C. 4	D. 5

42. Which alignment can produce a solar eclipse?

A. Sun - Earth - Moon	B. Moon - Sun - Earth
C. Sun - Moon - Earth	D. Earth - Sun - Moon

43. Which pair correctly matches Earth's motions and times?

A. rotation - 365 days; revolution - 24 hours	B. rotation - one month; revolution - one week
C. rotation - one year; revolution - one day	D. rotation - about 24 hours; revolution - about 365 days

44. Kabir checks a cyclone map and then follows a route to a shelter. Which two satellite uses are involved?

A. weather observation and navigation	B. communication and Moon exploration
C. tides and rocket staging	D. constellations and simple machines

45. A rocket has used all the fuel in its first stage. What is the best next action in a multistage design?

A. Carry the empty stage forever	B. Separate the empty stage
C. Turn off every remaining engine	D. Open the astronaut's helmet

46. The four daily star counts are 14, 10, 16 and 12. What is their total?

A. 50	B. 54
C. 52	D. 16

47. Which statement correctly compares a star, a planet and a moon?

A. All three make their own light	B. A planet orbits a moon
C. A moon is always larger than a star	D. A star makes light; a planet orbits a star; a moon orbits a planet

48. The sequence new Moon -> first quarter -> full Moon takes about:

A. two weeks	B. one hour
C. one day	D. one year

49. Two children push a box with equal force in opposite directions. What is most likely if the box was at rest?

A. It moves quickly in both directions	B. It remains at rest
C. It flies upward	D. It loses gravity

50. All planets in our Solar System orbit the Sun. Earth is a planet. Which conclusion follows?

A. Earth is the Sun	B. The Sun orbits Earth every day
C. Earth orbits the Sun	D. Earth is a natural satellite

Mock Paper 5

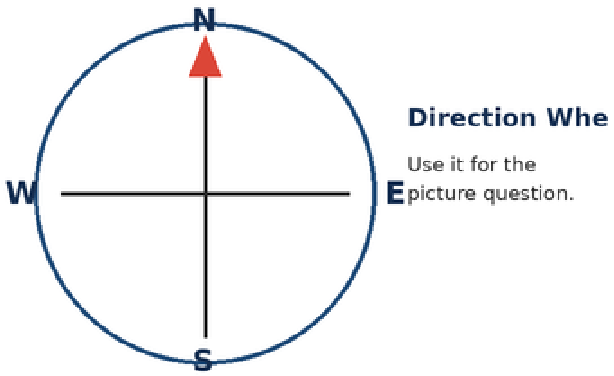
2025 Mock Questions - Grade 4 - Foundation Level

Questions	Marks	Time	Negative marking
50	50	75 minutes	None

Instructions: Choose one correct option for each question. Questions 41-50 are the Olympiad Challenge. Work carefully and use the pictures where shown.

Section A - Core Questions

1. On the direction wheel, which direction lies opposite East?



A. North	B. South
C. Up	D. West

2. A good astronomical observation record should include:

A. date, time and what was seen	B. only the observer's favourite colour
C. only the name of a planet	D. a guess with no observation

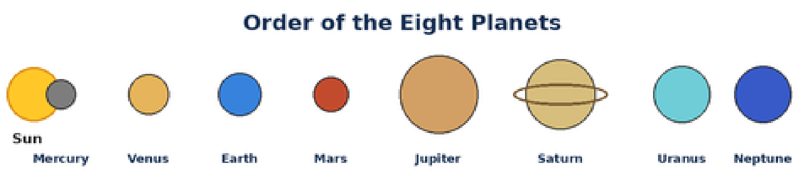
3. Earth has seasons mainly because it:

A. moves closer to the Moon every winter	B. revolves around the Sun with a tilted axis
C. stops rotating in summer	D. changes its size each month

4. Constellations help sky-watchers to:

A. make stars move into new places	B. turn planets into stars
C. recognise patterns and regions in the night sky	D. measure ocean depth directly

5. Which list gives the first four planets in the correct order from the Sun?



A. Venus, Mercury, Mars, Earth	B. Earth, Venus, Mercury, Mars
C. Mars, Earth, Venus, Mercury	D. Mercury, Venus, Earth, Mars

6. Which pair contains two rocky inner planets?

A. Earth and Mars	B. Jupiter and Saturn
C. Saturn and Neptune	D. Jupiter and Uranus

7. Pluto is classified as a:

A. star	B. dwarf planet
C. natural satellite of Earth	D. comet tail

8. Which statement best separates a comet from an asteroid?

A. A comet is always a planet	B. An asteroid is made only of water
C. A comet contains much ice and can form a tail near the Sun	D. An asteroid shines by making its own light

9. Which planet is nearest to the Sun?

A. Venus	B. Earth
C. Mars	D. Mercury

10. The Sun is important to Earth because it provides:

A. light and heat	B. rings and craters
C. ocean salt only	D. all of Earth's rocks

11. Which statement is correct?

A. The Milky Way contains the whole Universe	B. The Universe contains many galaxies
C. Every star is a galaxy	D. Earth is larger than every galaxy

12. Two stars may look different in brightness because they can differ in:

A. number of oceans	B. amount of soil
C. distance and actual brightness	D. type of rocket stage

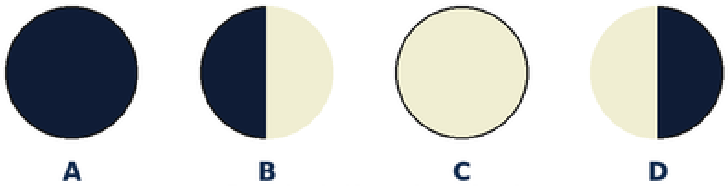
13. About how long does Earth take to rotate once?

A. one week	B. one month
C. 365 days	D. 24 hours

14. About how long does Earth take to revolve once around the Sun?

A. 365 days	B. 24 hours
C. seven days	D. one lunar night

15. In the Moon drawings, which phase normally comes after B as the lit part grows?



Four simple Moon-phase drawings

A. A	B. C
C. B again	D. D

16. During a lunar eclipse, which object is between the Sun and Moon?

A. Mars	B. Venus
C. Earth	D. a comet

17. Ocean tides are mainly linked with the pull of the:

A. North Star	B. Milky Way
C. weather satellite	D. Moon

18. A satellite that carries television and phone signals is a:

A. communication satellite	B. weather balloon only
C. natural satellite	D. dwarf planet

19. A weather satellite is most useful for observing:

A. the inside of a school bag	B. clouds, storms and large weather systems
C. underground train seats	D. the colour of one leaf indoors

20. GPS satellites help a navigation app find:

A. the age of the Sun	B. the taste of water
C. location and route	D. the number of stars in the Universe

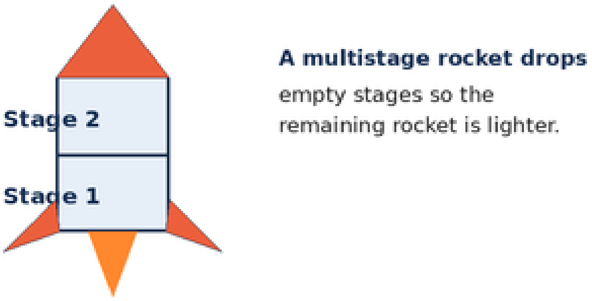
21. Which pair contains one natural and one artificial satellite?

A. Earth and Mars	B. Sun and Jupiter
C. comet and asteroid	D. Moon and weather satellite

22. A rocket rises because it pushes hot gases downward and the gases push the rocket:

A. upward	B. into the launch pad
C. towards Earth's centre only	D. without any motion

23. Why do many launch vehicles use stages?



A. Stages make gravity disappear	B. Empty parts can be dropped to reduce mass
C. Each stage becomes a planet	D. Stages stop the engine from working

24. A reusable launch vehicle is designed to:

A. remain on the Moon forever	B. turn into a constellation
C. use some major parts again	D. stop after one second

25. The full form of ISRO is:

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26. The Vikram lander and Pragyan rover were parts of:

A. Chandrayaan-3	B. Mangalyaan
C. Aditya-L1	D. Gaganyaan

27. India's Mars Orbiter Mission is popularly called:

A. Chandrayaan	B. Mangalyaan
C. Gaganyaan	D. Aryabhata only

28. Aditya-L1 is an Indian mission designed to observe the:

A. deep ocean	B. rings of Saturn only
C. Sun	D. surface of Mars

29. Gaganyaan aims to develop India's ability for:

A. moving the Moon	B. building a city on the Sun
C. stopping Earth's revolution	D. human spaceflight

30. Astronauts inside an orbiting spacecraft appear to float because they and the spacecraft are:

A. falling around Earth together	B. outside all gravity
C. being pushed by sunlight only	D. standing on a hidden floor

31. The International Space Station travels mainly in orbit around:

A. Mars	B. Earth
C. the Sun at L1	D. Jupiter

32. A space telescope can observe from above much of Earth's:

A. core	B. ocean floor
C. atmosphere	D. soil

33. If a moving toy car is pushed harder in the same direction, it will usually:

A. turn into a magnet	B. lose all mass
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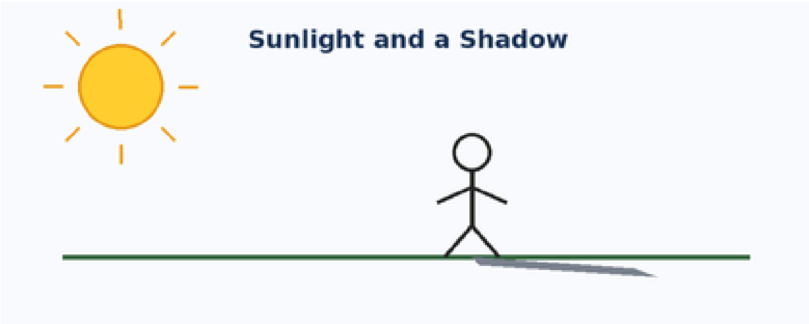


A. an opaque object blocks light	B. sound passes through air
C. heat reaches water	D. gravity pulls a stone

35. Which object changes electrical energy mainly into light?

A. a wooden ruler	B. an electric bulb
C. a stone	D. a paper clip alone

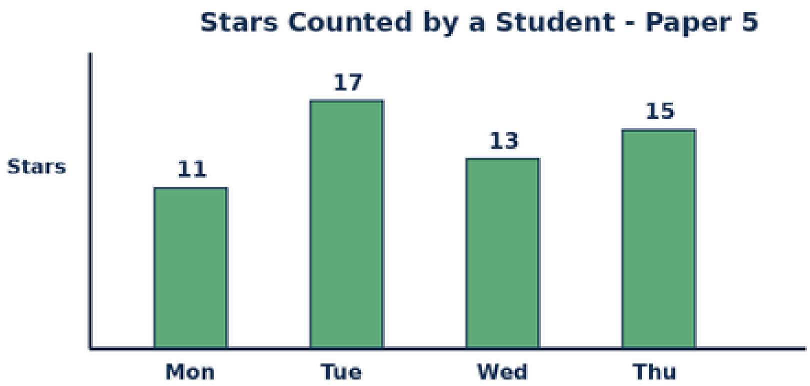
36. A ramp is an example of which simple machine?

A. pulley	B. wheel only
C. inclined plane	D. thermometer

37. Earth's atmosphere is important because it provides air and also helps:

A. stop all sunlight	B. remove gravity
C. make Earth square	D. protect life and control temperature

38. Use the bar chart. What is the difference between the highest and lowest star counts?



A. 6	B. 7
C. 17	D. 56

39. Complete the number pattern: 7, 9, 11, ___.

A. 12	B. 13
C. 15	D. 11

40. Which mission-purpose match is correct?

A. Mangalyaan - study of the Sun	B. Aditya-L1 - human flight around Earth
C. Chandrayaan - Moon exploration	D. Gaganyaan - study of Saturn's rings

Section B - Olympiad Challenge

These ten questions are harder, but every question remains within the Grade-level Foundation syllabus.

41. Using the planet order, how many planets lie between Venus and Saturn?

A. 2	B. 4
C. 5	D. 3

42. Which alignment can produce a solar eclipse?

A. Sun - Moon - Earth	B. Sun - Earth - Moon
C. Moon - Sun - Earth	D. Earth - Sun - Moon

43. Which pair correctly matches Earth's motions and times?

A. rotation - 365 days; revolution - 24 hours	B. rotation - about 24 hours; revolution - about 365 days
C. rotation - one month; revolution - one week	D. rotation - one year; revolution - one day

44. Vihaan checks a cyclone map and then follows a route to a shelter. Which two satellite uses are involved?

A. communication and Moon exploration	B. tides and rocket staging
C. weather observation and navigation	D. constellations and simple machines

45. A rocket has used all the fuel in its first stage. What is the best next action in a multistage design?

A. Carry the empty stage forever	B. Turn off every remaining engine
C. Open the astronaut's helmet	D. Separate the empty stage

46. The four daily star counts are 11, 17, 13 and 15. What is their total?

A. 56	B. 54
C. 58	D. 17

47. Which statement correctly compares a star, a planet and a moon?

A. All three make their own light	B. A star makes light; a planet orbits a star; a moon orbits a planet
C. A planet orbits a moon	D. A moon is always larger than a star

48. The sequence new Moon -> first quarter -> full Moon takes about:

A. one hour	B. one day
C. two weeks	D. one year

49. Two children push a box with equal force in opposite directions. What is most likely if the box was at rest?

A. It moves quickly in both directions	B. It flies upward
C. It loses gravity	D. It remains at rest

50. All planets in our Solar System orbit the Sun. Earth is a planet. Which conclusion follows?

A. Earth orbits the Sun	B. Earth is the Sun
C. The Sun orbits Earth every day	D. Earth is a natural satellite

Answer Keys

Each table gives the correct option letter. Challenge explanations follow the key for each paper.

Mock Paper 1

1. D	2. A	3. B	4. C	5. D	6. A	7. B	8. C	9. D	10. A
11. B	12. C	13. D	14. A	15. B	16. C	17. D	18. A	19. B	20. C
21. D	22. A	23. B	24. C	25. D	26. A	27. B	28. C	29. D	30. A
31. B	32. C	33. D	34. A	35. B	36. C	37. D	38. A	39. B	40. C
41. D	42. A	43. B	44. C	45. D	46. A	47. B	48. C	49. D	50. A

Challenge Explanations - Paper 1

41. D - Write the planets in order and count only those between Venus and Jupiter.

42. A - A solar eclipse occurs when the Moon is between the Sun and Earth.

43. B - Earth rotates once in about a day and revolves around the Sun once in about a year.

44. C - Weather satellites watch storms, while navigation satellites support route finding.

45. D - Separating the empty stage reduces mass and improves efficiency.

46. A - Add all four values: 8 + 12 + 9 + 14 = 43.

47. B - Stars produce light, planets orbit stars, and natural satellites orbit planets.

48. C - The Moon takes about two weeks to move from new Moon to full Moon.

49. D - Equal opposite forces balance, so there is no net force to start the box moving.

50. A - This is a simple logical conclusion from the two given statements.

Mock Paper 2

1. B	2. C	3. D	4. A	5. B	6. C	7. D	8. A	9. B	10. C
11. D	12. A	13. B	14. C	15. D	16. A	17. B	18. C	19. D	20. A
21. B	22. C	23. D	24. A	25. B	26. C	27. D	28. A	29. B	30. C
31. D	32. A	33. B	34. C	35. D	36. A	37. B	38. C	39. D	40. A
41. B	42. C	43. D	44. A	45. B	46. C	47. D	48. A	49. B	50. C

Challenge Explanations - Paper 2

41. B - Write the planets in order and count only those between Mercury and Mars.

42. C - A solar eclipse occurs when the Moon is between the Sun and Earth.

43. D - Earth rotates once in about a day and revolves around the Sun once in about a year.

44. A - Weather satellites watch storms, while navigation satellites support route finding.

45. B - Separating the empty stage reduces mass and improves efficiency.

46. C - Add all four values: 10 + 7 + 13 + 11 = 41.

47. D - Stars produce light, planets orbit stars, and natural satellites orbit planets.

48. A - The Moon takes about two weeks to move from new Moon to full Moon.

49. B - Equal opposite forces balance, so there is no net force to start the box moving.

50. C - This is a simple logical conclusion from the two given statements.

Mock Paper 3

1. D	2. A	3. B	4. C	5. D	6. A	7. B	8. C	9. D	10. A
11. B	12. C	13. D	14. A	15. B	16. C	17. D	18. A	19. B	20. C
21. D	22. A	23. B	24. C	25. D	26. A	27. B	28. C	29. D	30. A
31. B	32. C	33. D	34. A	35. B	36. C	37. D	38. A	39. B	40. C
41. D	42. A	43. B	44. C	45. D	46. A	47. B	48. C	49. D	50. A

Challenge Explanations - Paper 3

41. D - Write the planets in order and count only those between Earth and Saturn.

42. A - A solar eclipse occurs when the Moon is between the Sun and Earth.

43. B - Earth rotates once in about a day and revolves around the Sun once in about a year.

44. C - Weather satellites watch storms, while navigation satellites support route finding.

45. D - Separating the empty stage reduces mass and improves efficiency.

46. A - Add all four values: 9 + 15 + 12 + 8 = 44.

47. B - Stars produce light, planets orbit stars, and natural satellites orbit planets.

48. C - The Moon takes about two weeks to move from new Moon to full Moon.

49. D - Equal opposite forces balance, so there is no net force to start the box moving.

50. A - This is a simple logical conclusion from the two given statements.

Mock Paper 4

1. B	2. C	3. D	4. A	5. B	6. C	7. D	8. A	9. B	10. C
11. D	12. A	13. B	14. C	15. D	16. A	17. B	18. C	19. D	20. A
21. B	22. C	23. D	24. A	25. B	26. C	27. D	28. A	29. B	30. C
31. D	32. A	33. B	34. C	35. D	36. A	37. B	38. C	39. D	40. A
41. B	42. C	43. D	44. A	45. B	46. C	47. D	48. A	49. B	50. C

Challenge Explanations - Paper 4

41. B - Write the planets in order and count only those between Mars and Uranus.

42. C - A solar eclipse occurs when the Moon is between the Sun and Earth.

43. D - Earth rotates once in about a day and revolves around the Sun once in about a year.

44. A - Weather satellites watch storms, while navigation satellites support route finding.

45. B - Separating the empty stage reduces mass and improves efficiency.

46. C - Add all four values: 14 + 10 + 16 + 12 = 52.

47. D - Stars produce light, planets orbit stars, and natural satellites orbit planets.

48. A - The Moon takes about two weeks to move from new Moon to full Moon.

49. B - Equal opposite forces balance, so there is no net force to start the box moving.

50. C - This is a simple logical conclusion from the two given statements.

Mock Paper 5

1. D	2. A	3. B	4. C	5. D	6. A	7. B	8. C	9. D	10. A
11. B	12. C	13. D	14. A	15. B	16. C	17. D	18. A	19. B	20. C
21. D	22. A	23. B	24. C	25. D	26. A	27. B	28. C	29. D	30. A
31. B	32. C	33. D	34. A	35. B	36. C	37. D	38. A	39. B	40. C
41. D	42. A	43. B	44. C	45. D	46. A	47. B	48. C	49. D	50. A

Challenge Explanations - Paper 5

41. D - Write the planets in order and count only those between Venus and Saturn.

42. A - A solar eclipse occurs when the Moon is between the Sun and Earth.

43. B - Earth rotates once in about a day and revolves around the Sun once in about a year.

44. C - Weather satellites watch storms, while navigation satellites support route finding.

45. D - Separating the empty stage reduces mass and improves efficiency.

46. A - Add all four values: 11 + 17 + 13 + 15 = 56.

47. B - Stars produce light, planets orbit stars, and natural satellites orbit planets.

48. C - The Moon takes about two weeks to move from new Moon to full Moon.

49. D - Equal opposite forces balance, so there is no net force to start the box moving.

50. A - This is a simple logical conclusion from the two given statements.

Parent and Teacher Review Guide

Score out of 50	Suggested interpretation	Next step
45-50	Very strong understanding	Review mistakes and try the next paper.
35-44	Good progress	Revise the units where errors occurred.
25-34	Developing understanding	Read the related textbook lessons and retry.
Below 25	Needs guided practice	Work slowly with a parent or teacher, one unit at a time.

How to Review an Incorrect Answer

Ask the child to explain the chosen option.

Return to the related concept instead of giving only the correct letter.

Use a simple drawing, model or observation when possible.

Let the child answer a similar question in their own words.

Praise careful thinking, not only a high score.